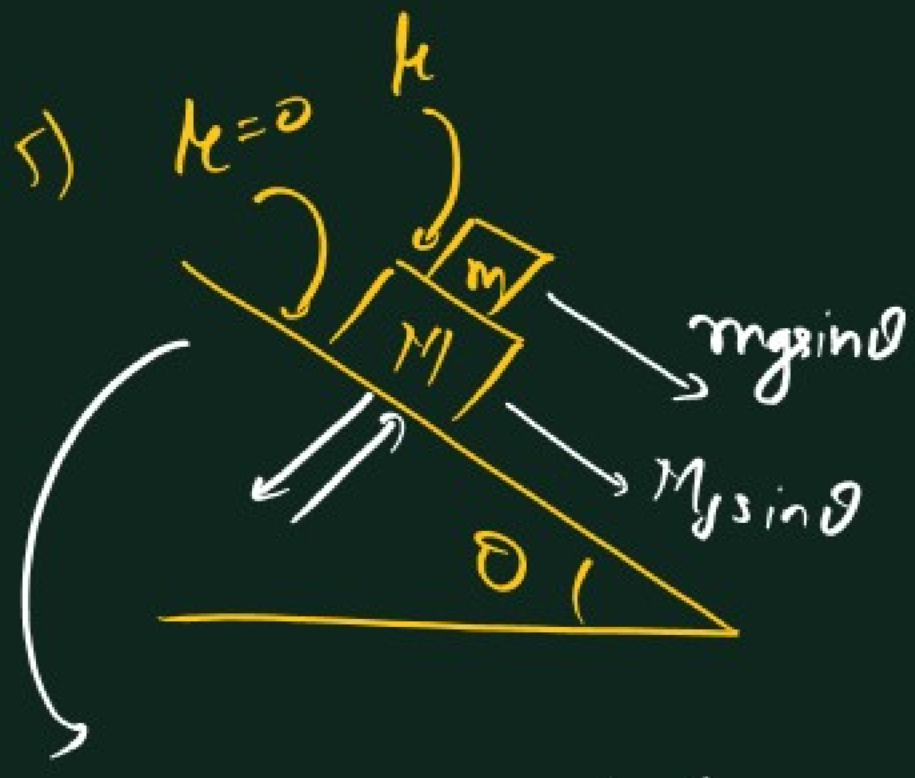
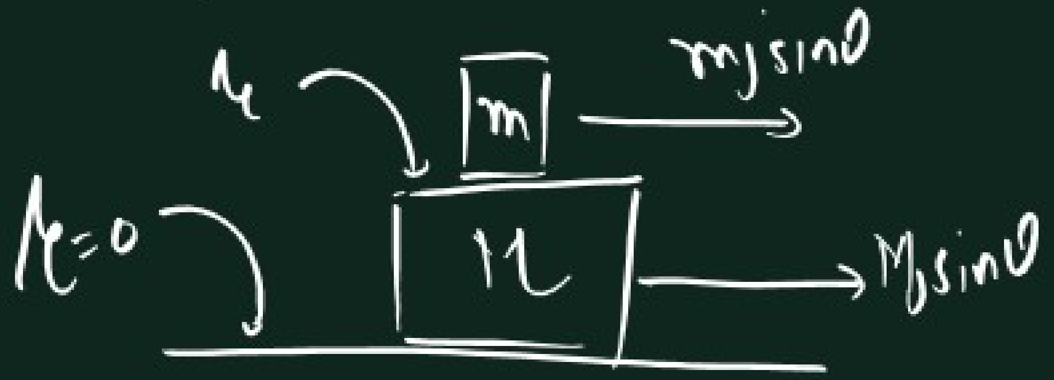
Doubts

25) 

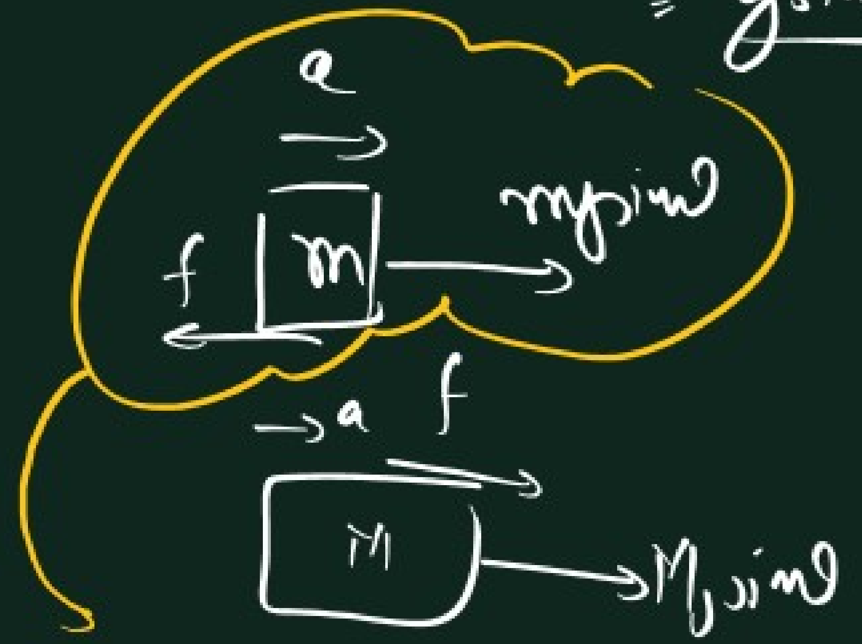


Step I: Assume સાથે

$$a = \frac{F_{net}}{m+M} = \frac{\cancel{(m+M)}\, g\sin\theta}{\cancel{m+M}}$$

$$= g\sin\theta$$

II



$$mg\sin\theta - f = ma \rightarrow g\sin\theta$$

$$\Rightarrow \cancel{mg\sin\theta} - f = \cancel{mg\sin\theta}$$

$$\Rightarrow \boxed{f = 0}$$

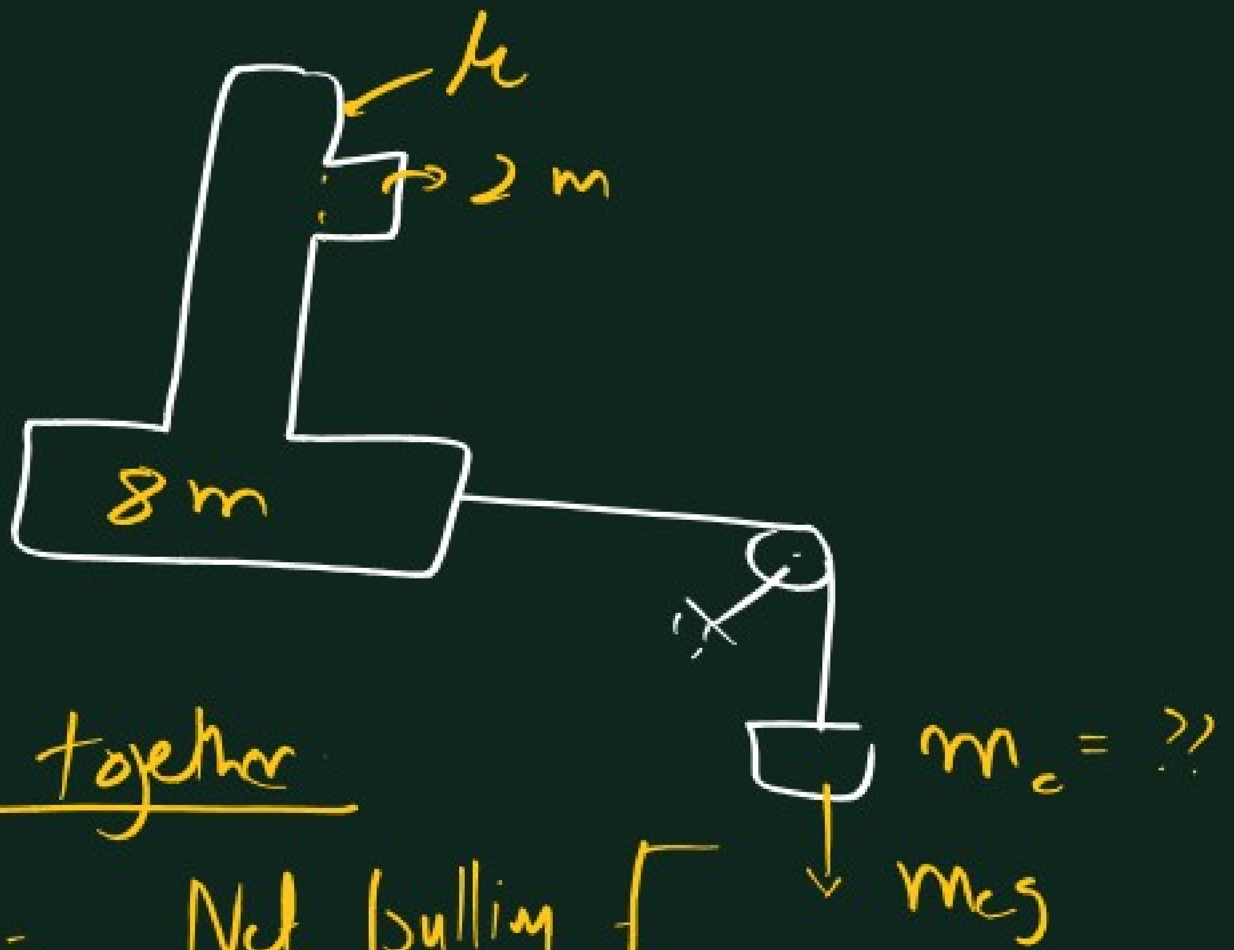
$\therefore$ Both move together.

$$a = g\sin\theta$$

$$f = 0$$

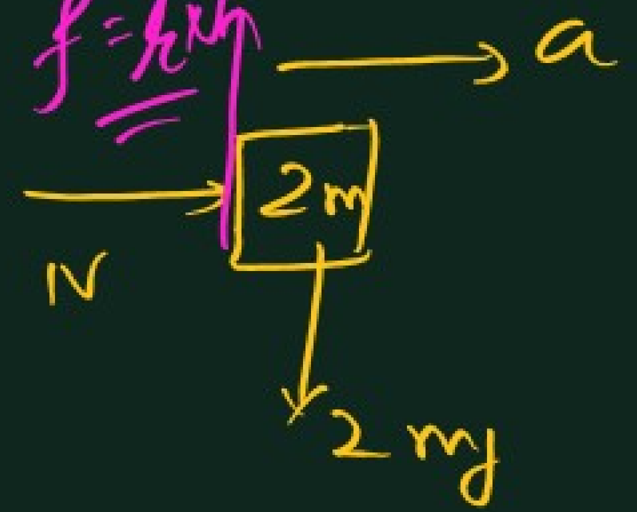
Ex 2

17)



All together

$$a = \frac{\text{Net pulling } f}{\text{total mass}} = \frac{m_c g}{10m + m_c} \quad \text{— (1)}$$



$$\Sigma F_x = ma$$

$$\Rightarrow N = (2m) \times \frac{m_c g}{10m + m_c} \quad \text{— (2)}$$

$$f = 2mg$$

$$\mu N$$

$$\Rightarrow \mu\left(\frac{\cancel{2m} m_c \cancel{g}}{10m + m_c}\right) = \cancel{2mg}$$

$$\Rightarrow \mu m_c = 10m + m_c$$

$$\Rightarrow m_c = \frac{10m}{\mu - 1}$$

1) $\boxed{\frac{d}{dx}(x^n) = n x^{n-1}}$

Eg. $x^2 = 2x^{2-1}$

$= \underline{\underline{2x}}$

$x^{3/2} = \frac{3}{2} x^{3/2 - 1}$

$= \frac{3}{2} x^{1/2} = \frac{3}{2}\sqrt{x}$

$\frac{d}{dx}\left(\frac{1}{x^2}\right) = \frac{d}{dx}(x^{-2})$

$= (-2) x^{(-2)-1}$

$= \underline{\underline{\frac{-2}{x^3}}}$

2) $\sin x \to \cos x$

$\cos x \to -\sin x$

$\tan x \to \sec^2 x$

$\sec x \to \tan x \sec x$

3) $e^x \to e^x$

$\ln x \to 1/x$

4) $C \to 0$

$(1, 2, \pi, e \ldots)$

5) $\frac{d}{dx}(u \pm v) = \frac{du}{dx} \pm \frac{dv}{dx}$

6) $\frac{d}{dx}(uv) = u\frac{dv}{dx} + v\frac{du}{dx}$

7) $\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{v(du/dx) - u\left(\frac{dv}{dx}\right)}{v^2}$

Q1. $\frac{d}{dx}(3e^x \sin x) = 3\left(e^x \times \frac{d\sin x}{dx} + \sin x \times \frac{de^x}{dx}\right)$

$= 3(e^x \cos x + \sin x e^x)$

$= \underline{3e^x(\cos x + \sin x)}$

Q2. $\frac{d}{dx}(x \ln x)$

$= \cancel{x} \times \frac{1}{\cancel{x}} + \ln x \times 1 = \underline{1 + \ln x}$

Q3. $\frac{d}{dx}(\underline{x} + \underline{2\ln x}) = 1 + 2 \times \frac{1}{x}$

$= \underline{1 + \frac{2}{x}}$

Chain rule

1) $\sin x = \cos x$

$$\boxed{\sin[f(x)] = \cos(f(x)) \times \frac{d f(x)}{dx}}$$

Eg. $\sin(x^2) = \cos(x^2) \times \frac{d(x^2)}{dx} \to 2x$

$= 2x\cos(x^2)$

Eg. $\sin(\ln x) = \cos(\ln x) \times \boxed{\frac{d(\ln x)}{dx}} \to \frac{1}{x}$

$= \frac{\cos(\ln x)}{x}$

Eg. $\frac{d}{dx}\sin(x^2+x)$

$= \cos(x^2+x) \times \boxed{\frac{d(x^2+x)}{dx}} \to (2x+1)$

$= (2x+1)\cos(x^2+x)$

Eg. $\sin(\underbrace{e^x+1}_{f(x)})$

$\frac{d}{dx}(\sin f(x)) = \cos f(x) \times \frac{d}{dx}(f(x)) \to (e^x+1)$

$= \cos(e^x+1) \times (e^x+0)$

$= e^x\cos(e^x+1)$

Eg. $\frac{d}{dx}\sin(\underbrace{4x^2+e^x}_{f(x)})$

$\cos f(x) \times \frac{d}{dx} f(x)$

$= \cos(4x^2+e^x) \times \frac{d(4x^2+e^x)}{dx} \to 8x+e^x$

$= (8x+e^x)\cos(4x^2+e^x)$

Eg. $\frac{d}{dx}\sin(\underbrace{2x}_{f(x)})$

$= \cos(2x) \times \boxed{\frac{d(2x)}{dx}} \to 2$

$= 2\cos(2x)$

2) $\cos x \to -\sin(x)$

$$\boxed{\cos f(x) \Rightarrow -\sin f(x) \times \frac{d}{dx} f(x)}$$

Eg. $\underline{\cos(\underbrace{4x^3}_{f(x)})} \longrightarrow -\sin f(x) \times \frac{d}{dx} f(x)$

$= \underline{-12x^2 \sin(4x^3)}$

3) $\tan\{f(x)\} \Rightarrow \sec^2\{f(x)\} \times \frac{d}{dx} f(x)$

4) $\sec f(x) = \tan f(x) \times \sec f(x) \times \frac{d}{dx} f(x)$

1) $\frac{d}{dx} \tan(\underline{\underline{x^2}})$

$= \sec^2(x^2) \times \frac{d}{dx}(x^2)$ $\to 2x$

$= 2x \sec^2(x^2)$

2) $\frac{d}{dx} \underline{\sec}(\underline{\underline{\ln x}})$

$= \tan(\ln x) \sec(\ln x) \times \frac{d}{dx}(\ln x)$ $\to \frac{1}{x}$

$= \underline{\frac{\tan(\ln x) \sec(\ln x)}{x}}$

5) $\frac{d}{dx} e^{f(x)}$

$e^{f(x)}$

$= e^{f(x)} \times \frac{d}{dx} f(x)$

6) $\frac{d}{dx}\{\ln f(x)\}$

$= \underline{\frac{1}{f(x)} \times \frac{d}{dx} f(x)}$

Eg. $\frac{d}{dx}\left(\ln(\underbrace{\sin x}_{f(x)})\right)$

$= \frac{1}{\sin x} \times \boxed{\frac{d}{dx}(\sin x)}$ $\to \cos x$

$= \frac{\cos x}{\sin x} = \underline{\cot(x)}$

7) Power chain rule.

$\rightarrow x^n = n x^{n-1}$

$\rightarrow \boxed{(f(x))^n = n(f(x))^{n-1} \times \frac{d}{dx}(f(x))}$

Q1) $\frac{d}{dx}(\underbrace{x^2+3}_{f(x)})^{4}$ ← $n=4$

$= 4(x^2+3)^{4-1} \times \boxed{\frac{d}{dx}(x^2+3)} \rightarrow 2x$

$= 8x(x^2+3)^3$

$\frac{d}{dx}(x^{1/2}) = \frac{1}{2}x^{1/2-1} = \frac{1}{2x^{1/2}}$

Q2 $\frac{d}{dx}(\underbrace{1+\ln x}_{f(x)})^2$

$= 2(1+\ln x)^{2-1} \times \boxed{\frac{d}{dx}(1+\ln x)} \rightarrow 0+\frac{1}{x}$

$= \frac{2(1+\ln x)}{x}$

Q3. $\frac{d}{dx}(\underbrace{1+e^x}_{f(x)})^3 = 3(1+e^x)^{3-1} \times \boxed{\frac{d}{dx}(1+e^x)} \rightarrow e^x$

$= 3e^x(1+e^x)^2$

Q4. $\frac{d}{dx}\sqrt{\underbrace{1+\sin x}_{f(x)}} = \frac{d}{dx}(f(x))^{1/2}$

$= \frac{1}{2}(1+\sin x)^{1/2-1} \times \boxed{\frac{d}{dx}(1+\sin x)} \rightarrow \cos x$

$= \frac{1}{2\sqrt{1+\sin x}} \times \cos x$

Q. $y = \sin x$

$\downarrow$

$\frac{dy}{dx} = \cos x$

$\downarrow$

$\frac{d^2y}{dx^2} = -\sin x$

$\downarrow$

$\frac{d^3y}{dx^3} = -\cos x$

$\vdots$

$$\left\{ \begin{array}{l} y \rightarrow \frac{dy}{dx} \rightarrow \frac{d^2y}{dx^2} \rightarrow \frac{d^3y}{dx^3} \cdots \\ y \rightarrow y' \rightarrow y'' \rightarrow y''' \cdots \end{array} \right.$$

Q. If $y = \ln x$, find 1) y''
2) y'''

$\downarrow$

$y' = \frac{1}{x}$

$\downarrow$

$y'' \rightarrow \frac{d}{dx}(1/x) = \frac{d}{dx}(x^{-1}) = -x^{-1-1} = -\frac{1}{x^2}$

$y''' = \frac{d}{dx}(-x^{-2}) = -(-2)x^{-2-1}$

$= \frac{2}{x^3}$

Q. If $y = x \ln x$
find y''

$\rightarrow \frac{dy}{dx}$ or $y' = \cancel{x} \times \frac{1}{\cancel{x}} + \ln x \times 1$

$= 1 + \ln x$

$\rightarrow y'' = 0 + \frac{1}{x} = 1/x$

Q. If $y = x^3 + 4x^2 + 2$

find y'' at $x=1$.

A- $y' = 3x^2 + 8x + 0$

$y'' = 6x + 8$

$\downarrow x = 1$

$\boxed{y'' = 14}$

Parametric form.

Eg. $y = 3t^2$
$x = t^3$

Find $\left(\frac{dy}{dx}\right)$

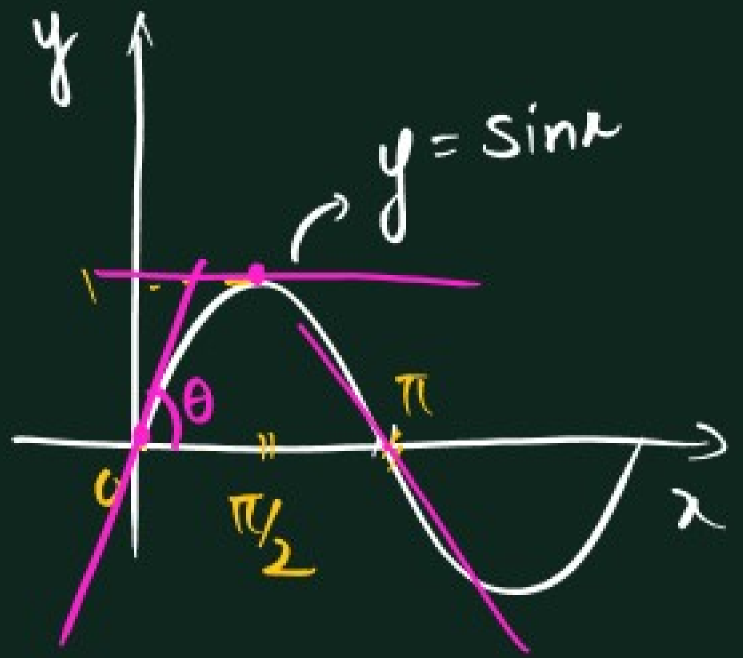
$$\boxed{\frac{dy}{dx} = \frac{(dy/dt)}{(dx/dt)}}$$

$$\frac{dy}{dt} = \frac{d}{dt}(3t^2) = 6t$$

$$\frac{dx}{dt} = \frac{d}{dt}(t^3) = 3t^2$$

$$\therefore \frac{dy}{dx} = \frac{6t}{3t^2} = \frac{2}{t}$$

graphical interpretation



$$\frac{dy}{dx} = \frac{d(\sin x)}{dx} = \boxed{\cos x}$$

$\Downarrow$

$$\boxed{\frac{dy}{dx} = \text{Slope of tangent at any point}}$$

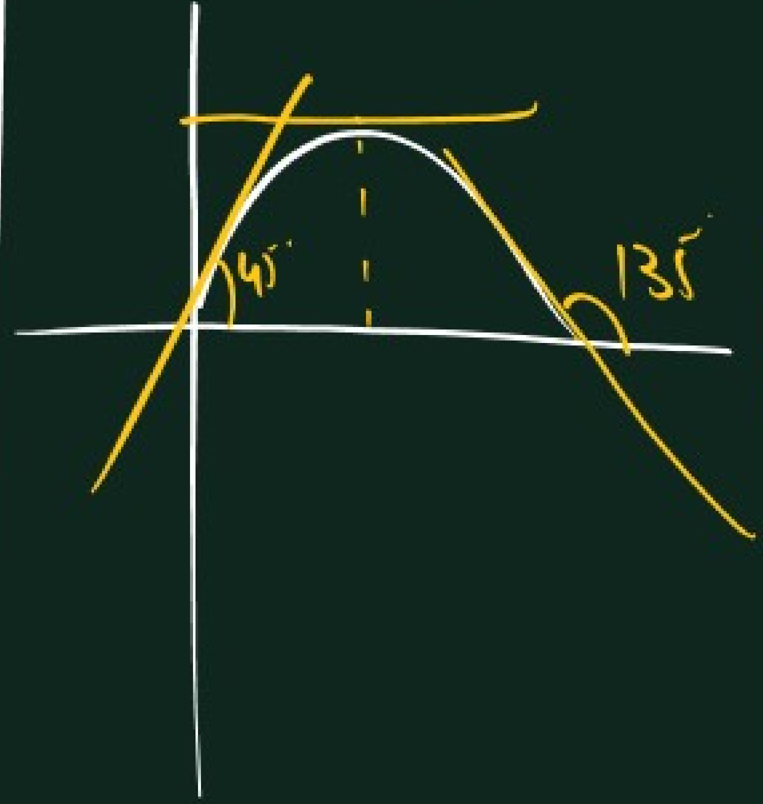
$x = 0$ $\quad \frac{dy}{dx} = \cos 0 = 1$

ie $\tan\theta = 1$ at $x = 0$

$\Rightarrow \theta = 45^\circ$

$\rightarrow$ $x = \pi/2$ $\quad \frac{dy}{dx} = \cos\frac{\pi}{2} = 0$

$\rightarrow$ $x = \pi$ $\quad \frac{dy}{dx} = \cos\pi = -1$



$(\tan(135^\circ) = -1)$

Q1) If $y = 3x + 1$

Find slope of the line

$$\frac{dy}{dx} = \frac{d(3x+1)}{dx}$$
$$= 3 \times 1 + 0 = 3$$

2) If $y = 2x^2 - 4x + 1$

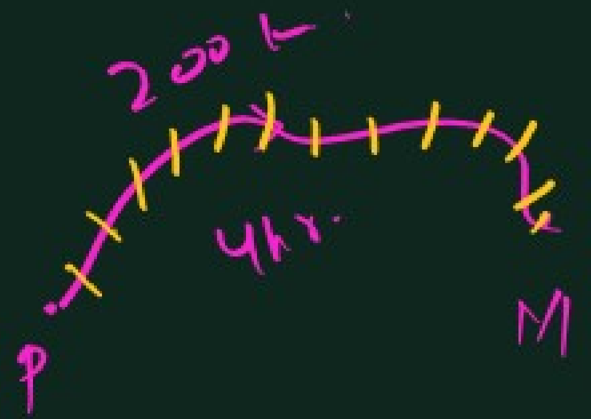
Find x where slope is 0

$$\frac{dy}{dx} = 4x - 4 + 0 = 0$$

$\Rightarrow 4x = 4$

$\Rightarrow \boxed{x = 1}$

Rate of change



→  200km in 4hrs

$$\frac{\text{Change in distance}}{\text{" " time}} = \frac{200}{4} = 50\ km/hr = \text{speed} = \text{R.O.C of distance with time.}$$

→ Avg. R.O.C vs. Instantaneous R.O.C.

Eg. $x = (t^2 + 1)$

t = seconds.

$t = 0,\ x = 1$

$t = 1,\ x = 2$

$t = 2,\ x = 5$

Find out Avg. R.O.C of x with time 't' from $t = 0$ to $t = 5$.

Ans $t = 0,\ x = 1$

$t = 5,\ x = 26.$

∴ R.O.C of x wrt t

$$= \frac{\text{Change in } x}{\text{" " } t} = \frac{x_2 - x_1}{t_2 - t_1} = \frac{26 - 1}{5 - 0} = \boxed{5\ m/s}$$

Find r.o.c of x with t

from $t = 2$ to $t = 2.5$

$(x = t^2 + 1)$

$t = 2, \; x = 2^2 + 1 = 5$

$t = 2.5, \; x = (2.5)^2 + 1 = 7.25$

$\therefore$ R.O.C $= \dfrac{x_2 - x_1}{t_2 - t_1} = \dfrac{7.25 - 5}{0.5}$

$= \dfrac{2.25}{0.5} = 4.5$ m/s

Find r.o.c. of x wrt time t

at $t = 2$ seconds.

$t = 2 \rightarrow x = 2^2 + 1 = 5$

$t = 2.1 \rightarrow x = (2.1)^2 + 1$

$= 4.41 + 1$

$= 5.41$

R.O.C $= \dfrac{5.41 - 5}{2.1 - 2}$

$= \dfrac{0.41}{0.1} = 4.1$ m/s

$2 - 2.0000\ldots 1$

Here $\dfrac{dx}{dt} = \dfrac{d}{dt}(t^2 + 1)$

$= 2t$

$t = 2$

$\dfrac{dx}{dt} = 4$ m/s

R.O.C of x with t at $t = 2$s.

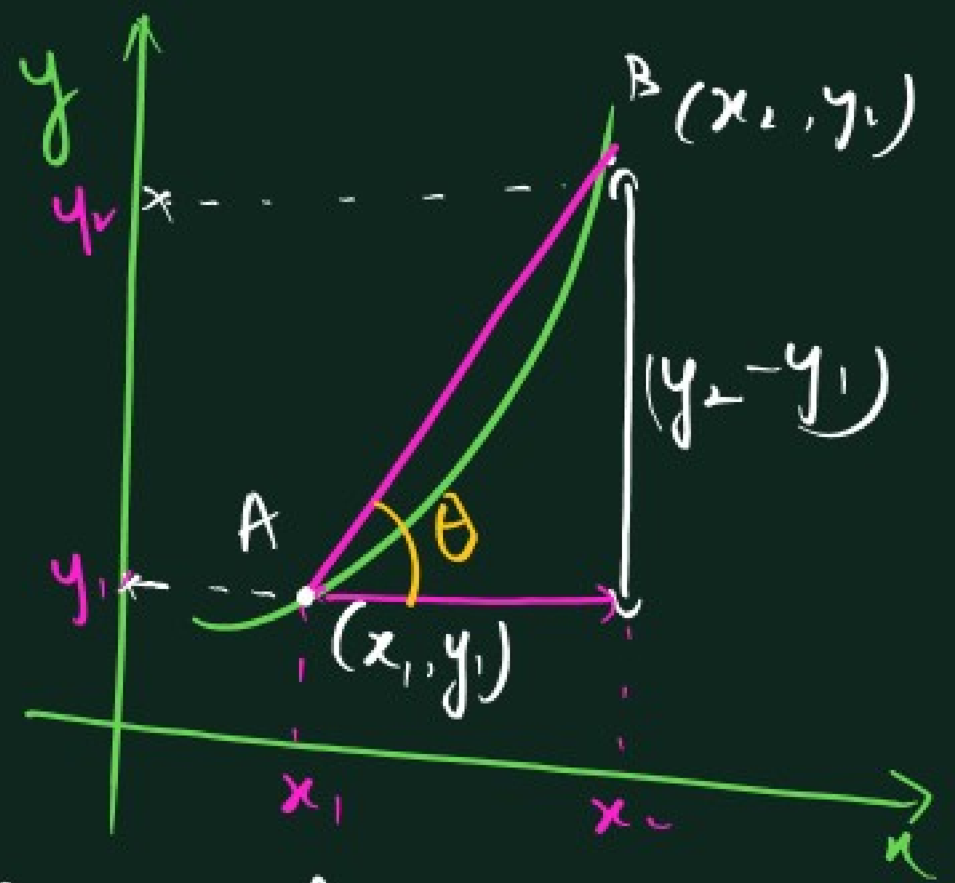
$$\boxed{\text{Average R.O.C} = \frac{\text{Finite change in } y}{\text{'' '' in } x} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\Delta y}{\Delta x}}$$

($\Delta \leftarrow$ change)

$$\frac{200}{4} = \underline{50 \text{ km/hr.}}$$

$$\boxed{\text{Inst. R.O.C} = \lim_{\Delta x \to 0} \left(\frac{\Delta y}{\Delta x}\right) = \frac{dy}{dx}}$$

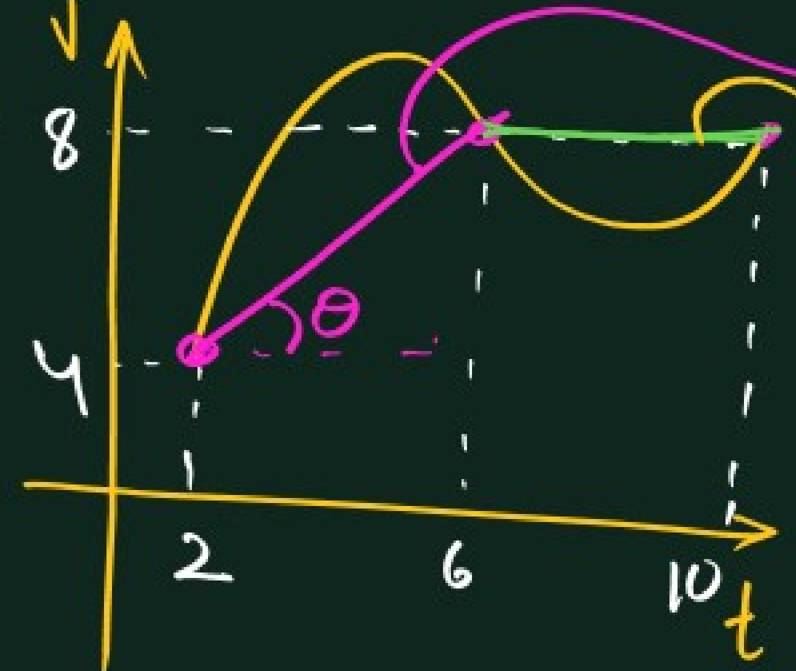
($dx \to$ very small change)



→ Avg. R.O.C from A to B $= \dfrac{\Delta y}{\Delta x} = \dfrac{y_2 - y_1}{x_2 - x_1}$

$\Rightarrow \boxed{\text{Avg. R.O.C} = \dfrac{\Delta y}{\Delta x} = \tan\theta}$

θ of line joining A, B

Q. 

Find avg. acceleration (r.o.c of v w.r.t. t) from

1) $t = 2$ to $t = 6$

2) $t = 6$ to $t = 10$

Slope $\tan\theta$ $\left(\text{or } \dfrac{y_2 - y_1}{x_2 - x_1}\right)$

$= \dfrac{8 - 4}{6 - 2}$

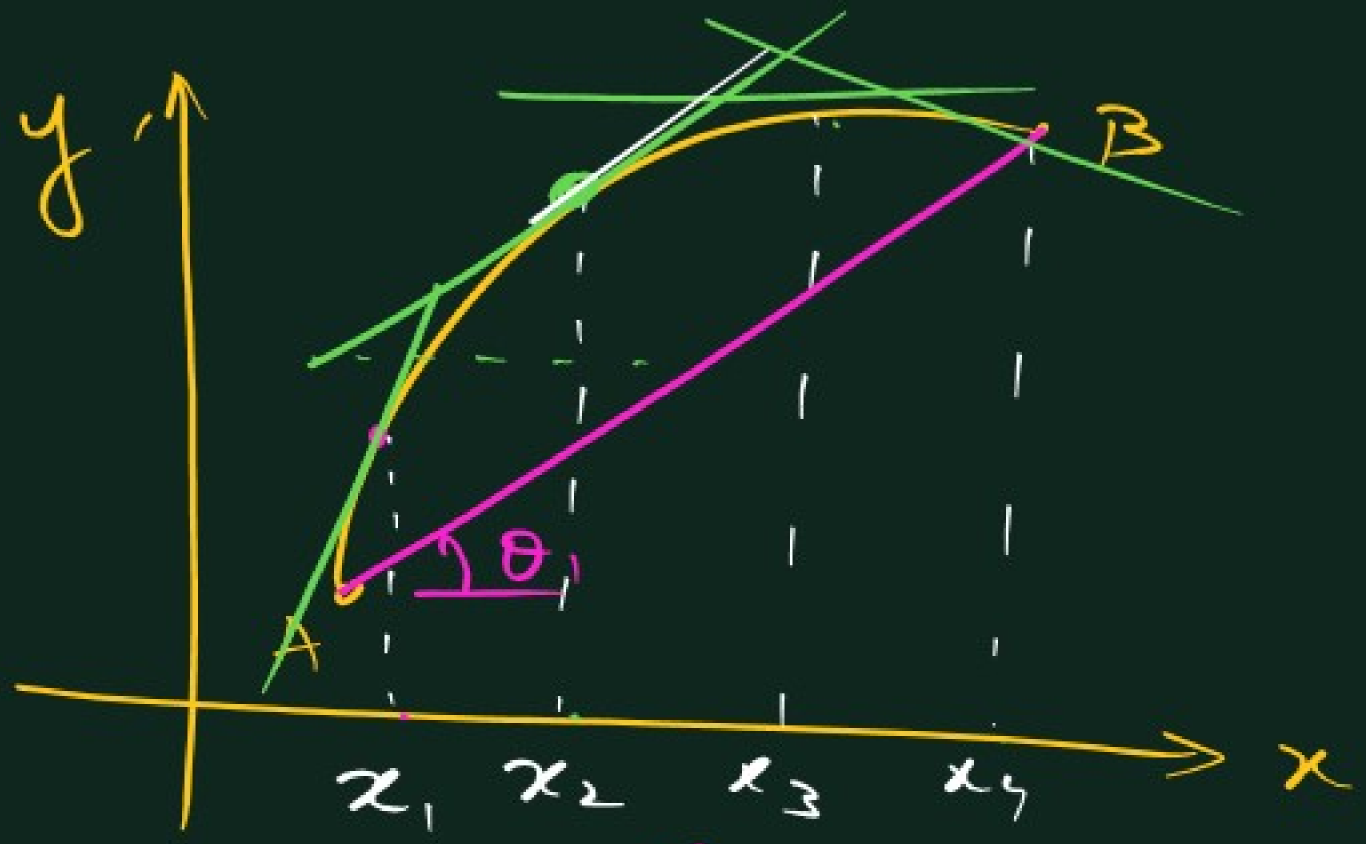
$= 4/4 = 1$

→ $t = 6$ to 10

Slope $= \dfrac{y_2 - y_1}{x_2 - x_1} = \dfrac{8 - 8}{10 - 6}$

$= 0$

Slope of horizontal line $= 0$



A) x_1 ~~B)~~ x_2

C) x_3 D) x_4

$$\text{Avg. R.O.C}_{A\to B} = \tan\theta_1$$

$\therefore$ For x_2 slope of tangent same as Avg. R.O.C $_{A\to B}$.

Q. For $y = 3x^2 - 5$, find slope of tangent at $(2, 7)$.

x y

$$\boxed{\text{Slope of tangent} = \text{Inst. R.O.C} = \frac{dy}{dx}}$$

$\therefore \dfrac{dy}{dx} = 6x - 0 = 6x$

$x = 2$

$$\boxed{\frac{dy}{dx} = 12}$$

Eg. $\begin{cases} v = \dfrac{dx}{dt} \\ a = \dfrac{dv}{dt} \\ F = \dfrac{dp}{dt} \end{cases}$

$$x = \left(2t^3 - \frac{3}{2}t^2 + 4\right)$$

Find out 1) initial v & a

$t = 0$

2) time when $v = 0$.

$$v = \frac{dx}{dt} = 6t^2 - \frac{3}{\cancel{2}} \times \cancel{2}t$$
$$= 6t^2 - 3t$$

$$a = \frac{dv}{dt} = 12t - 3.$$

1) $t = 0$ $\quad v = 6 \times 0 - 3 \times 0 = 0$

$a = 12 \times 0 - 3 = -3$

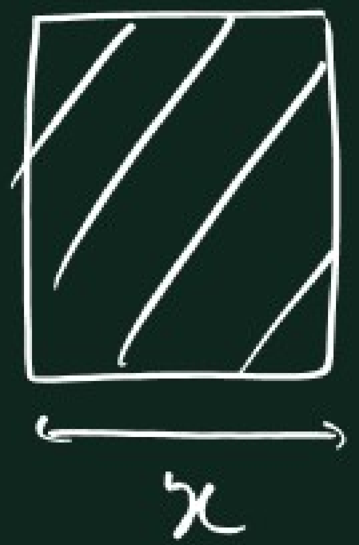
2) $v = 6t^2 - 3t$

$= 3t(2t - 1) = 0$

$\boxed{t = 0}$ or $2t - 1 = 0$

$\Rightarrow \boxed{t = 1/2}$

0.1 m/sec.



Given

R.O.C of x wrt $t = 0.1\ m/s$

$\Rightarrow \boxed{\dfrac{dx}{dt} = 0.1\ m/s}$

To find

$\dfrac{d(A)}{dt} = ??$ at $x = 2m$.

$A = x^2$

$\therefore \dfrac{d(x^2)}{dt} = ??$

$$\# \quad \boxed{\frac{d(\cancel{x^2})^{y}}{dt} = \frac{d(\cancel{x^2})^{y}}{dx} \times \frac{dx}{dt}}$$

Parametric

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$$

$$\Rightarrow \boxed{\frac{dy}{dt} = \frac{dy}{dx} \times \frac{dx}{dt}}$$

$$\frac{d(x^2)}{dt} = \frac{d(x^2)}{dx} \times \frac{dx}{dt}$$

$= 2x \times 0.1$

$= 0.2x$

For $x = 2m$

$\therefore \dfrac{dA}{dt} = 0.2 \times 2$

$= 0.4\ m^2/s.$

Q. given R.O.C $r' = 2$ cm/s

Find R.O.C of area

when $r = 14$ cm.

$\rightarrow \dfrac{dr}{dt} = 2$ cm/s

$\rightarrow$ To find $\dfrac{d(A)}{dt}$ at $r = 14$ cm, $A \rightarrow \pi r^2$

$$= \frac{d(\pi r^2)}{dt}$$

$$= \pi \frac{d(r^2)}{dr} \times \frac{dr}{dt}$$

$$= \pi \times 2r \times \frac{dr}{dt} = \frac{22}{7} \times 2 \times 14 \times 2 = 176 \text{ cm}^2/\text{s}$$

Q. At what radius will the rate of change of volume of sphere is same as rate of change of its radius.

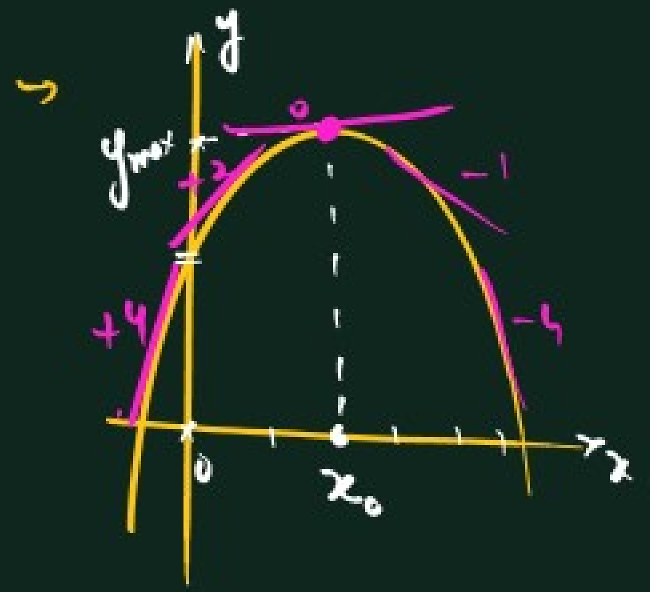
$$\frac{dr}{dt} = \frac{d(V)}{dt}, \quad V \rightarrow \left(\frac{4}{3}\pi r^3\right)$$

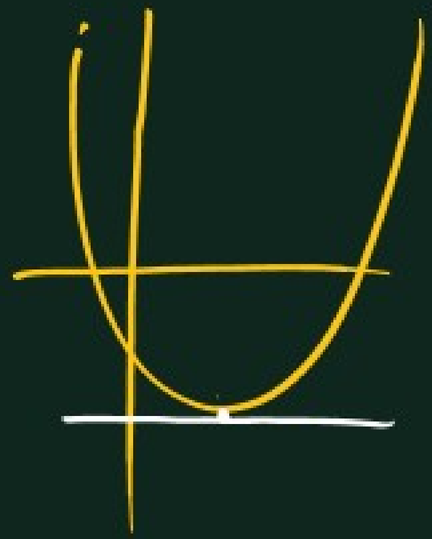
$$= \frac{4}{3}\pi \boxed{\frac{d(r^3)}{dt}}$$

$$\rightarrow \frac{d(r^3)}{dr} \times \frac{dr}{dt}$$

$$\Rightarrow \frac{d\cancel{r}}{\cancel{dt}} = \frac{4}{\cancel{3}}\pi \times \cancel{3}r^2 \times \frac{d\cancel{r}}{\cancel{dt}}$$

$$\Rightarrow r^2 = \frac{1}{4\pi} \Rightarrow \boxed{r = \frac{1}{\sqrt{4\pi}}}$$





1) $\left(\frac{dy}{dx}\right) = 0$

2) If $\frac{d^2y}{dx^2}$ or $y'' < 0 \Rightarrow$ Maxima

If $y'' > 0 \Rightarrow$ Minima.

Q1. $y = -x^2 + 6x + 4$

Find out

1) 'x' at which y is maximum/minimum? $\rightarrow x = 3$

2) max./min. value of y

3) Draw approx. graph.

1) $\frac{dy}{dx} = -2x + 6 = 0$

$\Rightarrow \boxed{x = 3}$

2) $\frac{d^2y}{dx^2} = -2 < 0$

$\Rightarrow x = 3$ is point of maxima.

Put $x = 3$ in y.

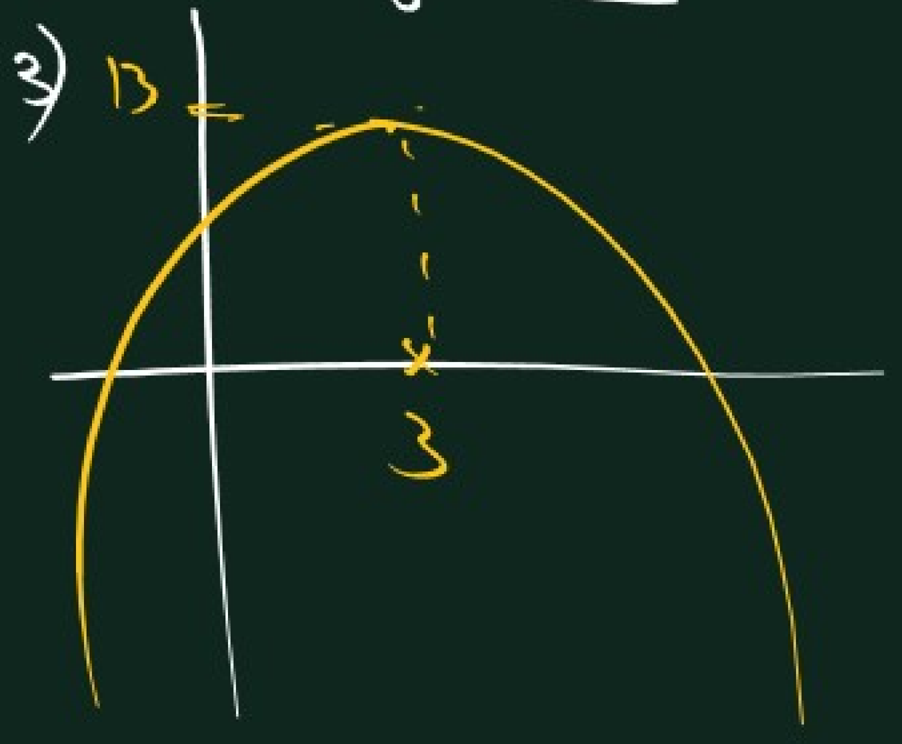
$\therefore y = -3^2 + 6 \times 3 + 4$

$= -9 + 18 + 4$

$= 13$

$\therefore$ Max. value of $y = 13$.

3)



$y = x^3 - 3x^2 + 6$

$\downarrow$

$y' = 3x^2 - 6x$

$y' = 0 \Rightarrow 3x^2 - 6x = 0$

$3x(x-2) = 0$

$\boxed{x = 0,\ x = 2}$

$\downarrow$

$\boxed{y'' = 6x - 6}$

$x = 0$ $y'' = 6\times 0 - 6 = -6 < 0 \leftarrow x = 0 \rightarrow$ maxima

$x = 2$ $y'' = 6\times 2 - 6 = 6 > 0 \leftarrow x = 2 \rightarrow$ minima

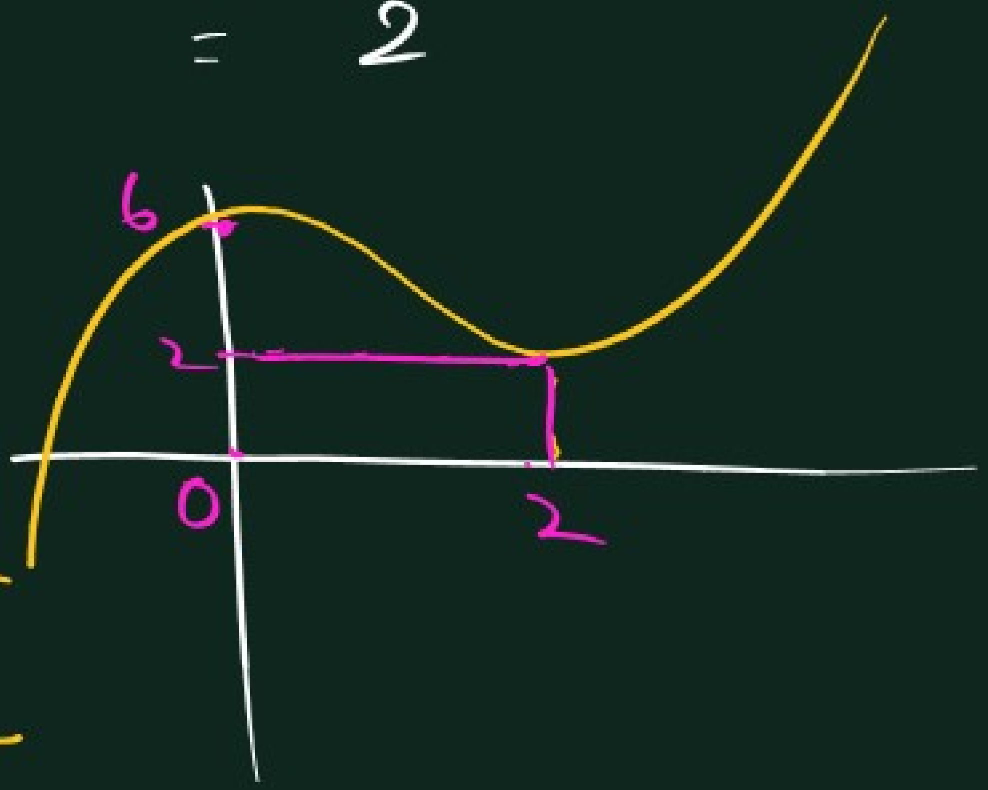
$\therefore$ $x = 0$ maxima.

$\therefore y_{max} = 0 - 3\times 0 + 6 = 6$

$x = 2 \rightarrow$ minima.

$y_{min} = 2^3 - 3\times 2^2 + 6$

$= 2$



Q. $y = 2x^3 - 24x + 2$

Find max. & min value of y.

Step 1 $\frac{dy}{dx} = \frac{d}{dx}(2x^3 - 24x + 2)$

$= 6x^2 - 24 = 0$

$\Rightarrow 6x^2 = 24$

$\Rightarrow x^2 = 4$

$\Rightarrow x = \pm 2$

Step 2 : Check for max./min.

$\frac{d^2y}{dx^2} = \frac{d}{dx}(6x^2 - 24) = 12x$

For $x = -2$ $\frac{d^2y}{dx^2} = -24 < 0$

$x = -2 \rightarrow$ point of max.

For $x = +2$ $\frac{d^2y}{dx^2} = +24 > 0$

$\Rightarrow x = +2$, point of min.

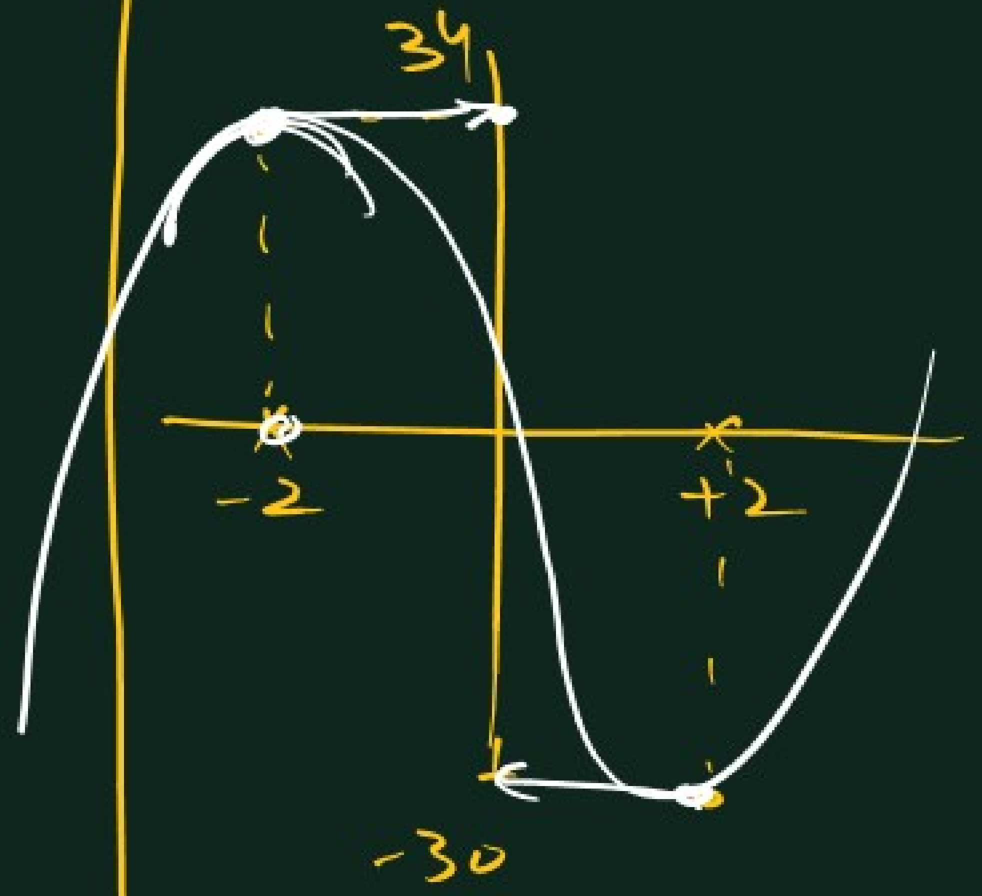
$\rightarrow$ To find max. value of $y \rightarrow$ put $x = -2$ in y

$\therefore y = 2(-2)^3 - 24(-2) + 2$

$= 34$

$y_{min} (x = +2)$ $y = 2 \times 2^3 - 24 \times 2 + 2$

$= -30$



(1) 3. If $y = \sin x$ & $x = 3t$ then $\frac{dy}{dt}$ will be $\frac{dy}{dt} = \frac{dy}{dx} \times \frac{dx}{dt}$

(a) $3 \cos (x)$ (b) $\cos x$

(c) $-3 \cos (x)$ (d) $-\cos x$

$= \cos x \times 3$

(2) 5. The displacement of a body at any time t after starting is given by $s = 10t - \frac{1}{2}(0.2)t^2$. The velocity of the body is zero after:

$10 - \frac{1}{2} \times 0.2 \times 2t = 0$

$v = \frac{ds}{dt}$

(a) 50 s (b) 100 s

(c) 80 s (d) 40 s

(3) 26. If $y = x^2 \sin x$, then $\frac{dy}{dx}$ will be …

$x^2 \cos x + 2x \sin x$

(4) 27. If $y = \tan x \cdot \cos^2 x$ then $\frac{dy}{dx}$ will be …

A) $1 - 2\sin^2 x$ B) $1 - 2\cos^2 x$

C) $1 - 2\tan^2 x$ D) $1 - 2\sec^2 x$

(5) 5. $f(x) = x^2 - 3x$, then the points at which $f(x) = f'(x)$ are

$f(x) = \frac{d f(x)}{dx}$

(a) 1, 3 (b) 1, –3

(c) –1, 3 (d) None of these

(6) 1. The value of the function $(x-1)(x-2)^2$ at its maxima is

(a) 1 (b) 2 (c) 0 (d) $\frac{4}{27}$

(7) 13. The function $f(x) = 2x^3 - 3x^2 - 12x + 4$ has

(a) No maxima and minima

(b) One maximum and one minimum

(c) Two maxima

(d) Two minima

(8) 15. The minimum value of $\left(x^2 + \frac{250}{x}\right)$ is

$y' = 2x - \frac{250}{x^2}$

$x^3 = 125$

$\Rightarrow x = 5$

(a) 75 (b) 50 (c) 25 (d) 55

$5^2 + \frac{250}{5} = 75$

32

4) $y = \tan x \cos^2 x$

$= \dfrac{\sin x}{\cancel{\cos x}} \times \cos^{\cancel{2}} x$

$y = \sin x \cos x$

$\downarrow$

$\dfrac{dy}{dx} = \sin x(-\sin x) + \cos x(\cos x)$

$= -\sin^2 x + \cos^2 x$

$\cos^2 x \to 1 - \sin^2 x$

$= 1 - 2\sin^2 x$

2nd way

$\dfrac{d}{dx}\left(\tan x \cos^2 x\right)$

$= \tan x \times \boxed{\dfrac{d}{dx}(\cos^2 x)} + \cos^2 x \dfrac{d}{dx}(\tan x)$

$\sec^2 x$

$\downarrow$

$= \dfrac{d}{dx}(\cos x)^2$

Power chain rule

$= 2 \times \cos x \times \dfrac{d}{dx}(\cos x)$

$= 2\cos x \times (-\sin x)$

$= \tan x \times 2\cancel{\cos x} \times (-\sin x) + \cancel{\cos^2 x} \times \dfrac{1}{\cancel{\cos^2 x}}$

$\tan x = \dfrac{\sin x}{\cancel{\cos x}}$

$= 1 - 2\sin^2 x$

5) $x^2 - 3x = 2x - 3$

$x^2 - 5x + 3 = 0$

6) $y = (x-1)(x-2)^2$

$= (x-1)(x^2-4x+4)$

$= (x^3 - 5x^2 + 8x - 4)$

$\frac{dy}{dx} = 3x^2 - 10x + 8 = 0$

$\Downarrow$

$\boxed{x = 4/3, 2}$

$\frac{d^2y}{dx^2} = 6x - 10$

$4/3 \;\; = -2$ — Maxima

$2 \;\; +2$ — Minima

$\therefore y_{max}$ at $(x = 4/3)$

$y_{max} = (4/3 - 1)(4/3 - 2)^2$

$= 1/3 \times \left(\frac{4}{9}\right) = 4/27$